Volume concept + cuboid/cuboid volume + surface area

Unit 1 • 3D shape graphics • 65 minutes • 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Unit 1 + Modern Education 5L B Unit 1

Core Traps: T1 Solid - Volume vs Surface Area Confusion • Unit Mismatch • Misuse of Cube Formula

SSPA Related: 🔴 High frequency The sub-test is required every year, accounting for about 12-15% of Paper 1

Prerequisite knowledge:P4 Perimeter and area of rectangle/square • Power calculation • Unit conversion (m↔cm)

Course goal:❶ Understand the concept and units of volume ❷ Formula for volume of cuboid ❸ Volume of cube ❹ Calculation of surface area and its difference with volume

Student Name: ______ Class: ______ Date: ______ Time Spent: ______

I.Warm-Up Questions（total 5 question，5 minutes）

#	Question	Difficulty	Working Space（Show full working）
1	The length of the rectangle is 8 cm, the width is 5 cm, the area = ? What is the unit?	Basic	
2	1 m = ？cm；1 m^2 = ？cm^2	Basic	
3	Calculate 12 × 5 × 3 = ?	Basic	
4	Calculate 7^3 = ? (i.e. 7×7×7)	Advanced	
5	How many faces does a cube have? What is the shape of each face?	Basic	

II.Core Knowledge + Worked Examples

Knowledge point 1: Volume concept and units (cm^3 and m^3) 🔴 SSPA

What is the volume?

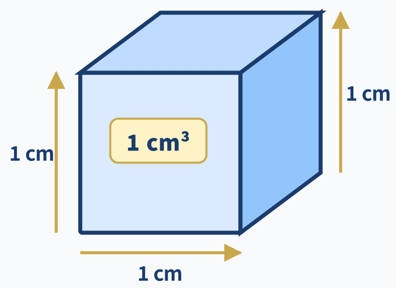
① Volume = object**The size of the space occupied**(three dimensions: length × width × height)

② 1 cm^3 = the space occupied by a cube with a side length of 1 cm (the size of a dice)

③ 1 m^3 = side length 1 m cube (about the size of a large washing machine)

④ **Unit conversion (very important!)**：1 m = 100 cm，1 m^3 = 100×100×100 = **1,000,000 cm^3**

⑤ **Area vs Volume |||SEP|||: Area = two dimensions (cm^2); volume = three dimensions (cm^3)** -Never confuse units!**Never mix up units!**



Cube with side length 1 cm → Volume = 1 cm^3 (size of three-dimensional space)

Trap detonation example

1 m^3 = ？cm^3

❌ Common mistakes (70% students)

1 m^3 = 100 cm^3 😅

I only did 1×100 and forgot to multiply each of the three dimensions by 100!

✅ Correct solution

1 m^3 = 100×100×100

= **1,000,000 cm^3**

1 m = 100 cm, cubed → 100^3

🧠 **Tip: "Multiply one dimension by 100, two dimensions by ten thousand, three dimensions by one million - convert m to cm and see clearly the power!"**

⚠ **The most frequent error: $m^3 \rightarrow cm^3$ forgetting to multiply by 1,000,000 (multiply the cubes by 100 each)**

⚠ **The second most frequent error: volume is measured in cm^2 - volume must be in "cubic" units (cm^3 / m^3)**

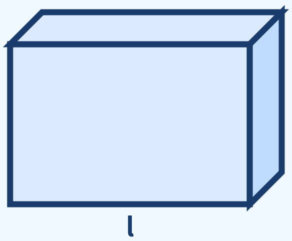
Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	Count: How many 1 cm^3 small cubes are made of in the picture below? (Picture: 2×3×2 cuboid)	🌱	
Example 2	2 m^3 = ? cm^3 (write down the conversion steps)	🌿	
Example 3	3,000,000 cm^3 = ? m^3	🌿	

Knowledge Point 1 Synchronization practice

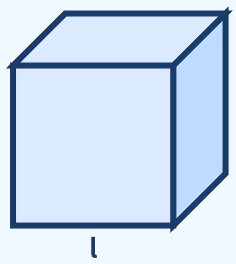
#	Question	Difficulty	Working Space
6	A cube with side length 1 cm has volume = ? What is the unit?	🌱	
7	1.5 m^3 = ? cm^3	🌿	
8	500,000 cm^3 = ? m^3 (answer as a decimal)	🌿	

📦 **Volume representation of cuboid: length × width × height (three-dimensional space)**



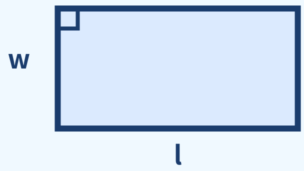
Length = 8 units, width = 5 units, height = 3 units. Volume = 8×5×3=120 cubic units. Note that volume is a "three-dimensional" concept (cm^3), not area (cm^2)!

🟦 Cube (special cuboid)



Length = width = height = 5cm. V=a^3=5^3=125 cm^3. The cube is a special case of the cuboid.

📏 Comparison: Area (2D · cm^2)



Area of rectangle = l × w (two dimensions). Don' t get confused: the unit of area is cm^2 and the unit of volume is cm^3!

Knowledge point 2: Volume of cuboid V = length × width × height 🔴 SSPA required test

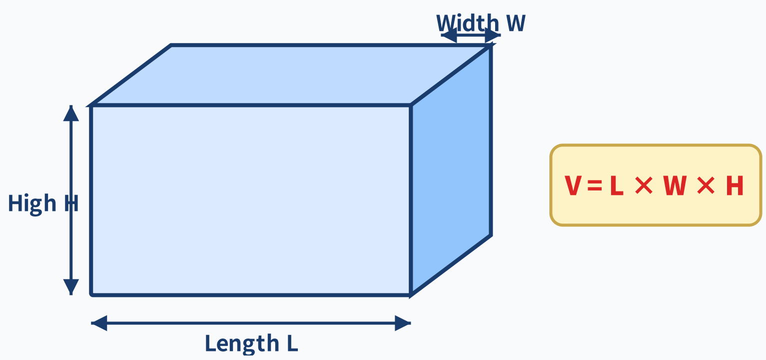
Cuboid volume formula:

① **V = L × W × H**(length × width × height)

② Essence: base area (length × width) × number of layers (height)

③ Extension: given V and two of the sides, the third side can be found → L = V ÷ W ÷ H

④ **Unit consistency**: Length, width and height must be in the same unit to be multiplied! Convert different units first



Volume of cuboid = length × width × height. Note that the three sides must have the same unit to be multiplied.

Trap detonation example

A cuboid is 2 m long, 50 cm wide, and 30 cm high. Volume = ? cm^3

❌ Common mistakes (60% students)

V = 2×50×30 = 3000 cm^3 ❌

The units are not unified! 2 m is 200 cm, not 2!

✅ Correct solution

2 m = 200 cm

V = 200×50×30

= **300,000 cm^3**

Unify the units first → then multiply!

🧠 Tip: "Length multiplied by width multiplied by height, the units must be consistent - m and cm cannot be mixed together and multiplied together!"

⚠️ Fatal error: Multiplying different units directly (2 m × 50 cm × 30 cm) → The answer must be wrong!

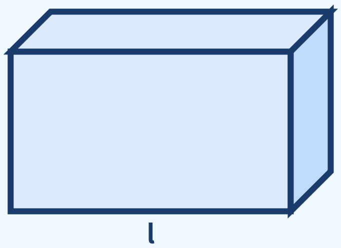
Knowledge Point 2 Worked Examples

#	Question	Difficulty	Working Space
Example 4	The cuboid is 10 cm long, 6 cm wide, and 4 cm high. Volume = ?	🌱	
Example 5	The cuboid is 1.5 m long, 0.8 m wide, and 0.5 m high. Volume = ? m^3	🌿	
Example 6	The cuboid has a volume of 240 cm^3, a length of 10 cm, and a width of 6 cm. High = ?	🌳	

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
9	Cuboid 12×5×8 cm. Volume = ?	🌱	
10	Cuboid 3 m × 2 m × 1.5 m. Volume = ? m^3	🌱	
11	The cuboid is 1 m long, 40 cm wide, and 25 cm high. Volume = ? cm^3	🌿	
12	The cuboid has a volume of 360 cm^3, a length of 12 cm, and a height of 5 cm. Broad = ?	🌿	
13	The cuboid is 80 cm long, 0.5 m wide, and 40 cm high. Volume = ? m^3 (answer as a decimal)	🌳	

📐 Volume of cuboid V = length × width × height — 3D visualization



Marking: length (l)=10cm, width (w)=6cm, height (h)=4cm. Volume=10×6×4=240 cm^3. The units of each side must be consistent!

Knowledge point 3: Volume of cube $V = a^3$ 🔴 SSPA required test

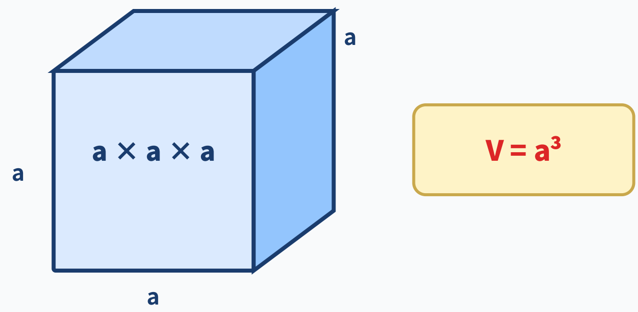
Special formula for cube:

① A cube is a cuboid with**length = width = height**→ V = a × a × a =$\mathbf{a^3}$

② a^3 means a multiplied by itself three times,**is not a×3**(This is the most common mistake!)

③ Example: side length 5 cm → V = 5^3 = 5×5×5 = 125 cm^3

④ Given the volume, find the side length: a = $\sqrt[3]{V}$ (open cube root, use trial calculation/factorization for P5-P6)



Cube = length = width = height = a, $V = a \times a \times a = a^3$. Note that $a^3 \neq a \times 3$!

□ **Trap detonation example**

The side length of a cube is 4 cm, its volume = ?

❌ Common mistakes (50% students)

$V = 4 \times 3 = 12\ cm^3$ ❌

Mistaking a^3 for $a \times 3$! A cube is multiplied by itself three times!

✅ Correct solution

$V = 4^3 = 4 \times 4 \times 4$

$= \mathbf{64\ cm^3}$

$a^3 = a \times a \times a$, not $a \times 3$

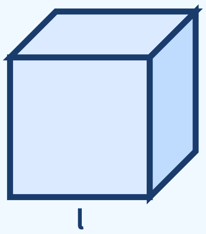
🧠 **Formula: "The volume of a square cubed, 4^3 is 64, not 12 - multiplied by itself three times, not three!"**

⚠ **The most fatal error: a^3 is treated as $a \times 3$. $4^3 = 64$, not 12. Write $4 \times 4 \times 4 = 64$ every time.**

Knowledge Point three example question + practice

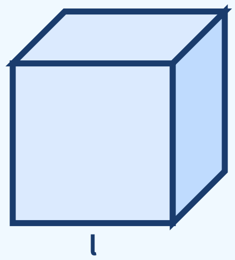
#	Question	Difficulty	Working Space
Example 7	The side length of a cube is 9 cm, its volume = ?	🌱	
Example 8	Volume of cube is 216 cm^3, side length = ?	🌳	
14	The side length of the cube is 7 cm. Volume = ?	🌱	
15	The side length of the cube is 0.5 m. Volume = ? m^3	🌿	
16	Cube volume 125 cm^3. Side length = ?	🌳	
17	The side length of the cube is 15 cm. Volume = ? cm^3; how much m^3?	🌳	

Cube 4cm



V=4^3=64 cm^3. Note: $4^3 \neq 4\times3$!

Cube 5cm



V=5^3=125 cm^3 ∘ SA=6×5^2=150 cm^2 ∘

⚠ **Note: $5^3 = 5\times5\times5 = 125$ (not 5×3=15). a^3 is a multiplied by itself three times, not a times 3!**

Knowledge Point 4: Surface Area (SA) VS Volume (V) 🔴 SSPA Advanced

Surface area formula:

① **Cuboid surface area**= 2×(LW + LH + WH) = 2×(Length×Width + Length×Height + Width×Height)

② **Cube surface area**= 6 × a^2 (6 identical square faces)

③ **SA unit is cm^2 (square meter) |||SEP|||, the unit of V is cm^3 (cubic meter)** -The unit is different!④ **Common test method: compare SA and V of two solids, or find the side length if SA is known**

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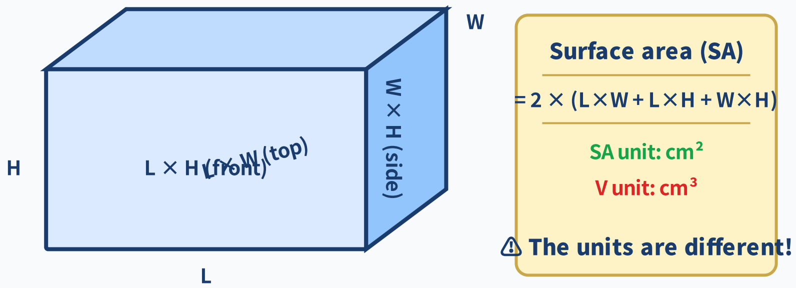


SA = sum of areas of 6 faces. Each side appears 2 times (front/back, up/down, left/right).

V	SA	L	⚠
volume	**surface area**	**total edge length**	**Don't be confused**
Length×width×height unit cm^3	2(LW+LH+WH) unit cm^2	4(L+W+H) unit cm	Three formulas and three units!

example

Example 9: cuboid 8×5×3 cm. Find (a) volume (b) surface area.

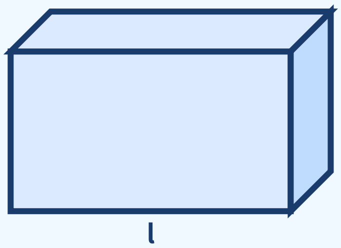
example

Example 10: The side length of the cube is 6 cm. Find (a) volume (b) surface area. Compare the values of V and SA.

Knowledge Point Four Synchronization Practice

#	Question	Difficulty	Working Space
18	Cuboid 10×6×4 cm. (a) Volume = ? (b) Surface area = ?	🌿	
19	The side length of the cube is 5 cm. (a) Volume = ? (b) Surface area = ?	🌱	
20	Cuboid 12×8×6 cm. Surface area = ?	🌿	

🔍 **Volume vs surface area - two ways to measure the same cuboid**



Volume (V) = internal space = 10 × 6 × 4 = 240 cm^3 (three dimensions · cubic). Surface area (SA) = outer wrapping paper = 2(10×6+10×4+6×4)=2(60+40+24)=248 cm^2 (two dimensions · square).

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
21	Cuboid 8×4×3 cm. Volume = ?	🌱	
22	The side length of the cube is 6 cm. Volume = ?	🌱	
23	Cuboid 5×5×10 cm. (a) Volume = ? (b) Surface area = ?	🌱	
24	The side length of the cube is 8 cm. Volume = ? Surface area = ?	🌿	
25	2.5 m^3 = ? cm^3	🌿	

Advanced layer (total 5 questions, 🧗🚀 choose do)

#	Question	Difficulty	Working Space
26	The cuboid is 2 m long, 1.2 m wide, and 80 cm high. Volume = ? m^3	🌿	
27	The cuboid has a volume of 480 cm^3, a length of 10 cm, and a width of 8 cm. High = ? Surface area = ?	🌳	
28	The surface area of the cube is 150 cm^2. Side length = ? Volume = ?	🌳	

29	Compare: A (cuboid 10×6×4 cm) and B (cube with sides 6 cm). Which one is larger? How much bigger?	🌳	
30	The cuboid is 1.5 m long, 0.8 m wide, and 60 cm high. Surface area = ? m^2	🌳	

🌳 challenge layer (total 3 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
31	The side length of a cube is twice the height of a cuboid. Cuboid 10×8×6 cm. How much larger is the volume of the cube than the volume of the cuboid?	🌳	
32	The internal dimensions of a rectangular water tank are 80 cm × 50 cm × 60 cm. (a) How many cm^3 of water can the tank hold? How many m^3 is it? (b) Internal surface area of the tank = ? (not counting the above)	🌳	
33	Assemble 8 cubes with sides 3 cm into one large cube. (a) Side length of a large cube = ? (b) Volume of large cube = ? (c) How much less is the SA of the large cube than the total SA of the 8 small cubes?	🧰	

IV.applicationquestionspecial topic (SSPA text question)

#	Question	Difficulty	Working Space (column → calculate → answer sentence)
34	A rectangular fish tank is 60 cm long, 30 cm wide, and 40 cm high. What is the volume of the fish tank in cm^3? How many m^3 is it?	🌱	
35	A cube gift box has a side length of 15 cm. To wrap all sides of the gift box with floral paper (without overlapping), what is the minimum cm^2 of floral paper needed?	🌿	
36	The internal length of a rectangular container is 6 m, the width is 2.5 m, and the height is 2.4 m. How many cubes with a side length of 1 m can be placed in a container?	🌿	
37	The rectangular pool is 8 m long, 5 m wide, and 1.5 m deep. The walls and bottom of the pool should be paved with tiles (not the top). What is the total area to be tiled in m^2?	🌳	
38	A square volume of wood has a side length of 10 cm. Cut it into 8 small cubes of the same size. (a) The side length of each small cube = ? (b) Volume of each small cube = ?	🌳	

39	A cuboid room is 5 m long, 4 m wide, and 3 m high. (a) Volume of room = ? m^3 (b) Total surface area of four walls and ceiling = ? m^2 (excluding floor)		

V. Ultimate challenge area

#	Question	Difficulty	Working Space
1	The volume of a cube is 512 cm^3. (a) Side length = ? (b) Surface area = ? (c) If this cube is cut into smaller cubes with a side length of 4 cm, how many cubes can be cut out?		
2	The cuboid is 12×9×8 cm and the side of the cube is 10 cm. (a) What is the difference in volume between the two? (b) What is the difference between the surface areas of the two? (c) If the height of a cuboid is increased by 2 cm, will the new volume be larger or smaller than the cube?		
3	A cuboid water tank without a lid is made of 1 cm thick glass. External dimensions are 52 cm × 32 cm × 42 cm. (a) Internal volume (capacity) of the water tank = ? (b) How many cm^2 of glass are needed to make this tank? (excluding top cover)		

VI.Class afterhomework

Basic must-do questions (total 5 questions)

#	Question	Difficulty	Working Space
H1	Cuboid 9×5×4 cm. Volume = ?		
H2	The side length of the cube is 7 cm. Volume = ? Surface area = ?		
H3	Cuboid 12×8×10 cm. (a) Volume = ? (b) Surface area = ?		
H4	3 m^3 = ? cm^3 (write down the conversion steps)		
H5	The cuboid has a volume of 360 cm^3, a length of 12 cm, and a height of 5 cm. Find the breadth and surface area.		

For advanced, choose doquestion (total 2 questions)

#	Question	Difficulty	Working Space
H6	The surface area of the cube is 294 cm^2. Side length = ? Volume = ?		

H7	A room is 6 m long, 4 m wide and 3 m high. How many m^2 are the total areas of the four walls? What is the minimum roll required for wallpapering (5 m^2 per roll)?	🌳	

VII. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point
☐ I can complete 🌱basic questions independently
☐ I can challenge 🌿advanced questions
☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall
☐ Solve 🌱basic questions independently (100% correct)
☐ Challenge 🌿Advanced questions (80%+ correct)
☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	**a^3 mistaking a×3 |||SEP|||: side length 5 cm → V=5×3=15 (wrong!)**: side length 5 cm → V=5×3=15 (wrong!)	$a^3 = a \times a \times a$ ∘ $5^3 = 5 \times 5 \times 5 = 125$ cm^3
2	**m^3 ↔ cm^3 Conversion error |||SEP|||: 1 m^3 = 100 cm^3 (wrong!)**: 1 m^3 = 100 cm^3 (wrong!)	1 m^3 = 100×100×100 = 1,000,000 cm^3 (cubed, times 100)
3	**Multiply different units directly**：2 m × 50 cm × 30 cm	Unify the unit first (full rotation cm or full rotation m) → then calculate
4	**Volume vs surface area confusion |||SEP|||: Asked SA Answered V (unit cm^3 vs cm^2)**:Ask SA Answered V (unit cm^3 vs cm^2)	See clearly what the question is asking! V=three dimensions (cm^3); SA=sum of six sides (cm^2)
5	**The cube SA formula is wrong |||SEP|||: SA = a^3 or SA = $4 \times a^2$**: SA = a^3 or SA = $4 \times a^2$	Cube SA = $6 \times a^2$ (six identical faces)
6	**Cuboid SA omitted a certain face |||SEP|||: only 4 faces were counted**: Only 4 faces are counted	SA = 2(LW + LH + WH), six faces are the same.
7	**Missing units in application questions |||SEP|||: only write numbers, do not write cm^3 / m^3, or do not write the answer sentence**: Only write numbers, do not write cm^3 / m^3, or do not write an answer sentence	The submission test requires complete answers + correct units, otherwise step points will be deducted